

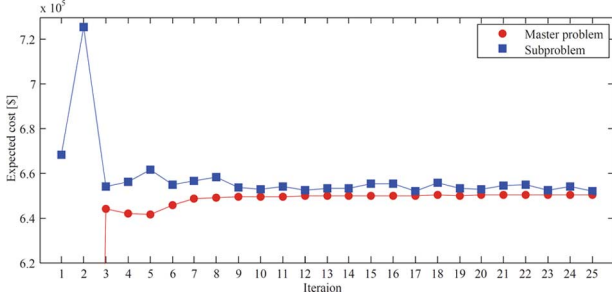


Fig. 5. Evolution of Benders' algorithm in Case B.

constraints (3j) render slightly different results. For example, the comparatively small generating unit 15 is fully committed in Case C, while it is fully decommitted in Case B. On the other hand, the comparatively large generating unit 9 is not committed in Case C for time periods t_1 and t_2 , while it is in Case B. These differences allow concluding that the voltage magnitude constraint may potentially alter the commitment results.

Finally, Fig. 5 illustrates the evolution of Benders' algorithm in Case B (ac-UC problem). This algorithm converges in iteration 25, where the difference between $Z_{up}^{(\nu)}$ (upper curve) and $Z_{down}^{(\nu)}$ (lower curve) is smaller than the considered tolerance.

D. Computational, Optimality, and Implementation Issues

Each subproblem (5) is solved using CONOPT [26] under GAMS [27], while the master problem (7) is solved using CPLEX 12.1 [25] under GAMS [27], both on a Sun Fire X4600M2 with 8 Quad-Core processors clocking at 2.9 GHz and 256 GB of RAM.

The following computational, optimality and implementation issues are in order:

- We report that we have verified the well-functioning of the heuristic technique explained in Section III-B. To this end, the expected cost for dc-UC problem is obtained using the proposed Benders' algorithm. Its difference with respect to that of Case A (the non-decomposed solution) is verified to be smaller than 0.2%.
- The computational times required for solving different instances are provided in the last column of Table VI. The computational time to solve Case B is comparatively higher. Note that both Cases B and C are solved iteratively, however the constraints of Case B are tighter. On the other hand, Case A is solved in a non-iterative way, thus its computational time is comparatively smaller.
- To speed-up convergence, suitable values for constants α^{down} , h^Q , and h^v need to be derived. In this paper, the values considered are -10^8 , 10^3 , and 10^3 , respectively, obtained through a trial-and-error analysis. Note that too large values of constants h^Q and h^v may lead to convergence fluctuations, while comparatively small values may lead to infeasible solutions.
- Note that the original ac-UC problem (1)–(3) is non-convex, and since the number of scenarios cannot be increased beyond a limit to ensure computational tractability, the result obtained may not be a global optimal. To overcome such issue, a multi-start decomposition

approach is used [28]. Accordingly, the proposed decomposition problem is solved several times considering diverse initial points, and finally the one leading to the smallest value for the objective function (1) is selected. Among the diverse initial solutions checked, the flat start, i.e., $\mathbf{v} = 1$ and $\boldsymbol{\theta} = 0$, shows the best performance.

- To assess the effect of the considered tolerance ε on the solution obtained, the proposed decomposition algorithm has been re-run considering a smaller value for ε , i.e., 0.1%. The total cost of operation has not significantly changed: \$651 304.6 for $\varepsilon = 0.1\%$ and \$651 909.9 for $\varepsilon = 0.3\%$ (Case B). However, the number of iterations required for the convergence of Benders' algorithm and therefore the CPU time increases significantly with $\varepsilon = 0.1\%$.
- Note that reducing the number of scenarios is computationally appropriate if decomposition techniques are not used, while a larger number of scenarios is generally better if decomposition techniques are used. It is important to note that using a larger number of scenarios generally results in higher accuracy.
- Note that in the decomposed cases considered (Cases B and C), the number of subproblems is 960 (40×24), being each of them comparatively small. The CPU time needed to solve each subproblem is about to 2% of the total time, which is reasonable. This means that the computational time may not significantly increase if a comparatively larger system is considered provided that the number of scenarios is reduced. However, to tackle case studies pertaining to real-world systems with thousands of nodes and lines and a large number of scenarios, the following additional alternatives are also available:
 - To use a supercomputer,
 - To implement parallelization techniques [29],
 - To apply appropriate techniques to simplify the network [30] and/or to reduce the number of scenarios [31],
 - To decompose the UC problem by area [32].

V. CONCLUSIONS

This paper proposes an efficient solution approach based on Benders' decomposition to solve a network-constrained ac-UC problem under uncertainty. The proposed approach allows decomposing such problem, which is mixed-integer nonlinear and generally intractable, into a mixed-integer linear master problem and a set of nonlinear but continuous subproblems, one per scenario and time period. Note that there is no off-the-shelf solver available for a MINLP problem, while MILP and NLP problems are both solvable using available commercial solvers. The numerical results obtained validate the well-functioning of the proposed approach.

We point out that the commitment status and dispatch results of generating units obtained by the dc and ac formulations might be different. The reasons for the differences are 1) the constraints that are not modeled in the dc-UC formulation, i.e., reactive power and voltage magnitude constraints, and 2) power losses which are not considered in the dc-UC formulation. It is also pointed out that the tightness level of voltage magnitude constraint in the ac-UC problem may potentially alter the commitment results.